

A standard bequest block: diagnosis of the estate-target failure and the M4 recalibration

Fertility–Housing project

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Executive conclusion

The July-15 attempt to estimate all three bequest parameters against three PSID estate moments failed for a reason that is now fully diagnosed: two of the three moments are not objects this model class can or should match. The dispersion ratio (p90/p50 of estate to income) requires an upper tail of financial wealth that no parameterization reaches — and in the data that tail is business, IRA, and other nonhousing wealth with no model counterpart. The family-size gap is statistically zero in the PSID (point estimate 0.101, bootstrap standard error 0.563) and is a knife-edge cancellation across marital strata that a model without marriage cannot represent. Because the dispersion moment carried an inverse-bootstrap-variance weight, it alone contributed 84% of the SMM loss, and chasing it degraded the eleven established moments by a factor of four relative to the July-15 mortality-only (M1) benchmark.

The repair is to return to the standard package used by the structural savings and housing-retirement literatures: a child-blind luxury warm glow (De Nardi, 2004; Lockwood, 2018; Nakajima and Telyukova, 2017), one estimated strength parameter, an externally anchored luxury shifter, and *level* targets for late-life wealth — the median estate-to-income ratio and the median nonhousing wealth-to-income ratio. The M4 recalibration implements exactly this. Headline result: [PENDING: M4 winner loss, estate median fit, nonhousing median fit, theta0 estimate vs zero-seed comparison].

1 Diagnosis: what the estate targets did

A matched PSID/model decomposition of the age 76–84 estate distribution (both normalized by each dataset’s median annual income at those ages) shows the failure is composition, not curvature:

Ages 76–84	PSID			Model (M3 winner)		
	p50	p75	p90	p50	p75	p90
Total estate	6.87	17.10	36.33	6.27	8.87	11.79
Nonhousing wealth	2.37	10.34	26.27	0.00	0.00	0.00
Home equity	3.19	6.40	11.61	6.27	8.87	11.79

The model matched the estate median almost exactly — with the wrong balance sheet. Housing supplies 99.9% of top-decile estate wealth in the model versus 18.6% in the PSID, and every old model household carries collateralized debt (liquid position $b < 0$ through the 90th percentile). The mechanism is transparent: after age 62 the model’s only borrowing is mortgage debt with a constant floor $b \geq -\phi pH$ and no amortization; the pension is a common scalar, so no late-life risk motivates liquid saving; and the warm glow values housing at gross price while a living sale nets $(1 - \psi)pH$, so dying in place dominates. Home equity is therefore the cheapest estate vehicle, and the model uses only it.

Critically, this is a property of the July-15 *calibration point*, not of the model. The M1 mortality-only winner — same constraints, no estimated bequest parameters — delivered a positive old nonhousing median of 2.00 against the PSID’s 2.23, while fitting the twelve shared non-estate moments at a summed loss of 3.84; the M3 winner fits the same twelve at 16.92. The unreachable dispersion row purchased nothing and paid with everything else, including flipping the young renter liquid-wealth moment to the wrong sign (−0.18 versus a target of +0.18, which M1 matched almost exactly).

2 The standard specification

M4 uses the bequest utility

$$B(W) = \theta_0 \frac{(\theta_1 + W)^{1-\sigma} - \theta_1^{1-\sigma}}{1-\sigma}, \quad W = \max\{b + pH, 0\}, \quad \sigma = 2, \quad (1)$$

where W is the household’s net worth at death: the liquid position (with mortgage debt as negative b) plus gross housing value, floored at zero. This is the luxury warm glow of De Nardi (2004); Lockwood (2018) estimates the same form, and Nakajima and Telyukova (2017) estimate it inside a housing-retirement model, which makes their calibration the closest anchor for this project.

External restrictions. The shifter is set at $\theta_1 = 0.25$ of average annual household income, anchored to the Nakajima and Telyukova (2017) estimate (their shifter $\zeta \approx \$19,600$; [PENDING: verified values, units, and dollar year from the paper — verification pass in progress]), with a De Nardi (2004)-scale sensitivity cell. The child multiplier is set to zero: $\theta_n = 0$. Both are external because the July-15 exercise demonstrated, constructively, that the available estate moments cannot identify them — the reachability frontier shows no θ_1 reconciles the median with the dispersion ratio, and the local Jacobian shows the θ_1 and θ_n columns are numerically indistinguishable.

Why a child-blind motive in a paper about children. Children in this model direct resources through housing while parents are alive: they raise required space, parents buy larger owner-occupied houses, transaction costs make those houses sticky into old age, and housing is the dominant component of estates. Parents therefore leave larger estates without any child-directed taste for posthumous giving. That is where the evidence points: Hurd (1989) finds the elderly with children decumulate no differently from the childless; Kopczuk and Lupton (2007) find children affect the *incidence* of a bequest motive, not its scale; De Nardi et al. (2025) find late-life saving behaves as luxury saving in which the number of children is not a conditioning variable; and even Kværner (2023), the strongest child-directed evidence, models the motive itself as child-blind, placing children in inter vivos transfers (cf. McGarry, 1999). Our own sample reproduces this null: the family-size gap in median estates at ages 65–75 is 0.101 with a bootstrap standard error of 0.563.

3 The M4 calibration

Fourteen moments, twelve estimated parameters (the eleven clean-frontier coordinates plus θ_0). The twelve established fertility, ownership, rooms, and young-liquid-wealth moments are unchanged. The two late-life wealth targets are levels, in the style of the median asset profiles targeted by De Nardi et al. (2010) and Nakajima and Telyukova (2017):

Moment	Target	Disciplines
Median total estate wealth / income, ages 76–84	6.501	θ_0
Median nonhousing wealth / income, ages 65–75	[PENDING: fresh bootstrap value]	late-life composition

Weights are inverse person-bootstrap variances (499 replications, person-level clustering). The estate dispersion ratio, the family-size estate gap, the housing/nonhousing decomposition, and the ownership path by age are reported as untargeted diagnostics. The search enforces nested-model dominance: the exact $\theta_0 = 0$ submodel (the M1 winner under the M4 objective) is injected as a seed, and the run fails if the free search ends above it.

4 Results

[PENDING: Full 14-moment target-fit table: moment, target, model, gap, weight, loss contribution.]

[PENDING: Full parameter table: 12 estimates with bounds and near-bound flags; external restrictions.]

[PENDING: Untargeted diagnostics: estate p90/p50, family-size gap, housing/nonhousing composition at 76–84, ownership path 62–78, comparison against the M1 and M3 columns.]

[PENDING: Nested-zero verdict: tight loss at the estimated θ_0 versus the $\theta_0 = 0$ seed; interpretation of whether the estate median lends support to a positive bequest strength.]

5 What remains open

The model still cannot produce the PSID estate tail, by design: that tail is financial and business wealth held by households with resources far above the pension-financed median, and matching it would require an earnings or entrepreneurial tail in the style of De Nardi (2004) or medical-expense risk in the style of De Nardi et al. (2010). Neither belongs in this paper unless upper-tail wealth becomes substantive for the housing–fertility question. The old-age leverage margin (no amortization of owner debt) also remains; policy counterfactuals that operate through late-life home equity should not be run off this block without further work.

References

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